

NYgrid 2019 interface flow

Compact-model calibration experiments

NPCC / New York modeling project

September 8, 2026

Project update

Models and comparison setup

Model	Buses	Role
Released NYgrid	57	Reproduce the paper's 2019 study
NY-only reference	46	Exact replay with boundary injections
Historical compact	51	Test matched 2019 inputs and Marcy paths
Year-end 2025 variant	71	Unchanged in this experiment

- Compare all **8,760 hours** using common 2019 inputs and paper-role interface definitions.
- Scale all compact-model power inputs and targets once by $\gamma = 0.35866$. Percentage errors are unchanged.

$$\text{WAPE} = 100 \frac{\sum |F_{\text{model}} - F_{\text{observed}}|}{\sum |F_{\text{observed}}|}$$

We report error relative to actual flow. The paper's Eq. 11 instead divides by the positive interface limit.

Source: [1–3]. Historical compact is a research reconstruction, not an exact 2019 as-built network.

NYgrid annual performance

Interface	WAPE (%)	MAE (MW)
Dysinger East	22.49	209.9
West Central	46.81	218.5
Total East	7.24	286.5
Moses South	8.99	152.7
Central East	9.63	211.8
UpNY-Coned	9.91	241.1
Dun/SPR-South	10.89	299.6

MAE uses actual-scale MW.

11.22% pooled WAPE

- Corrected five stale plotting formulas using the released interface map.
- Same solved states: pooled WAPE changes from 60.66% to 11.22%.
- West Central remains weakest, with 1,006 opposite-direction hours.

The formula correction repairs measurement; it changes no electrical parameters.

Source: [2]. Corrected paper-role operators, all 8,760 prepared hours. DC convergence alone does not establish AC or thermal feasibility.

How we adapted the paper's method

1. Transfer NYgrid's reconstructed generation, zonal load, and NY-side boundary injections to compact51.
2. Align bus allocation roles and interface cuts across the networks.
3. Test the paper-derived Marcy paths while retaining the inherited compact backbone.



Two added AC-network branches in the DC approximation, on a 100 MVA base. The model remains at 51 buses.

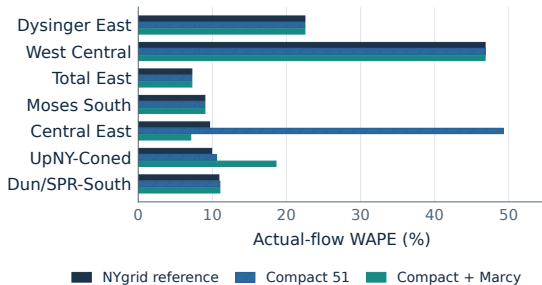
This is a calibration diagnostic: no new shared-parameter or per-hour target fitting.

Source: [1,3]. Source-informed reactances; AC parameters and ratings remain unqualified. Earlier compact S4/S7 calibration history is inherited.

Annual error improves, with a UPNY tradeoff

Annual interface error across networks

2019 | identical scaled inputs | paper-role operators



Pooled WAPE: 17.38% to 12.32%. Improvement appears in all 12 months.

Central East: 49.23% to 7.09%. **UPNY-ConEd: 10.55% to 18.59%.**

Source: [3]. Common paper-role cuts and 2019 inputs. Under current physical-zone cuts, pooled WAPE changes from 19.07% to 19.67%; operator choice matters.

Best examples: errors for every interface

2019 local hour	NYgrid reference Jan 8, 15:00	Compact + Marcy Jul 3, 04:00
Dysinger East	0.30%	1.46%
West Central	3.63%	1.48%
Moses South	4.07%	4.85%
Central East	1.73%	4.22%
Total East	2.49%	3.02%
UPNY-ConEd	2.22%	3.75%
Dunwoodie/Sprain	0.13%	2.45%
Pooled error	2.03%	3.28%
Worst interface	4.07%	4.85%

Each interface error is $100|F_{\text{model}} - F_{\text{observed}}|/|F_{\text{observed}}|$. These are different, retrospectively selected hours, with no opposite directions or imputed observations.

Source: [2,3]. Both are best minimax examples, ranked by the largest interface error. Selection is development evidence, not held-out validation.

What the results support and what comes next

A useful 2019 DC baseline and low-error compact examples

- **Main gaps:** West Central uncertainty and UPNY transfer sharing. Test bounded, shared allocation and network parameters.
- **AC qualification:** resolve Marcy line parameters, ratings, and controls. A separate three-hour AC checkpoint passed on compact51 *without Marcy*.
- **Independent evaluation:** freeze parameters before testing new periods. Use consistent 2025 data to validate the 71-bus expansion for DLR inference.

The selected 2019 snapshots do not establish accuracy of the 2025 model or AC/DLR feasibility of the Marcy variant.

Source: [3]. Next steps are proposed work. No new parameter fit or untouched predictive evaluation was completed in this experiment.

References

1. M. V. Liu et al. (2023). *An Open Source Representation for the NYS Electric Grid to Support Power Grid and Market Transition Studies*. IEEE Transactions on Power Systems, 38(4), 3293–3303.
[Paper DOI](#) [Pinned NYgrid source](#)
2. [NYgrid 2019 interface verification](#)
Annual results, formula audit, source-data checks, and selected hours.
3. [Matched compact 2019 comparison](#)
Network experiments, allocation tests, selected snapshots, and the separate AC checkpoint.

Scientific evidence: [project commit c472e25](#). Additional detail remains in the linked reports.